# User guide

How to load documents, ask questions in everyday words, and check how well the system works. Click any screenshot to open it full size.

## Getting started

This app answers questions from your own documents. You ask “Who appoints the Governor?” in plain language; it finds the matching passages even when the document uses different words, then writes a short answer with numbered citations. If the documents do not contain the answer, it says so instead of guessing.

### The six steps

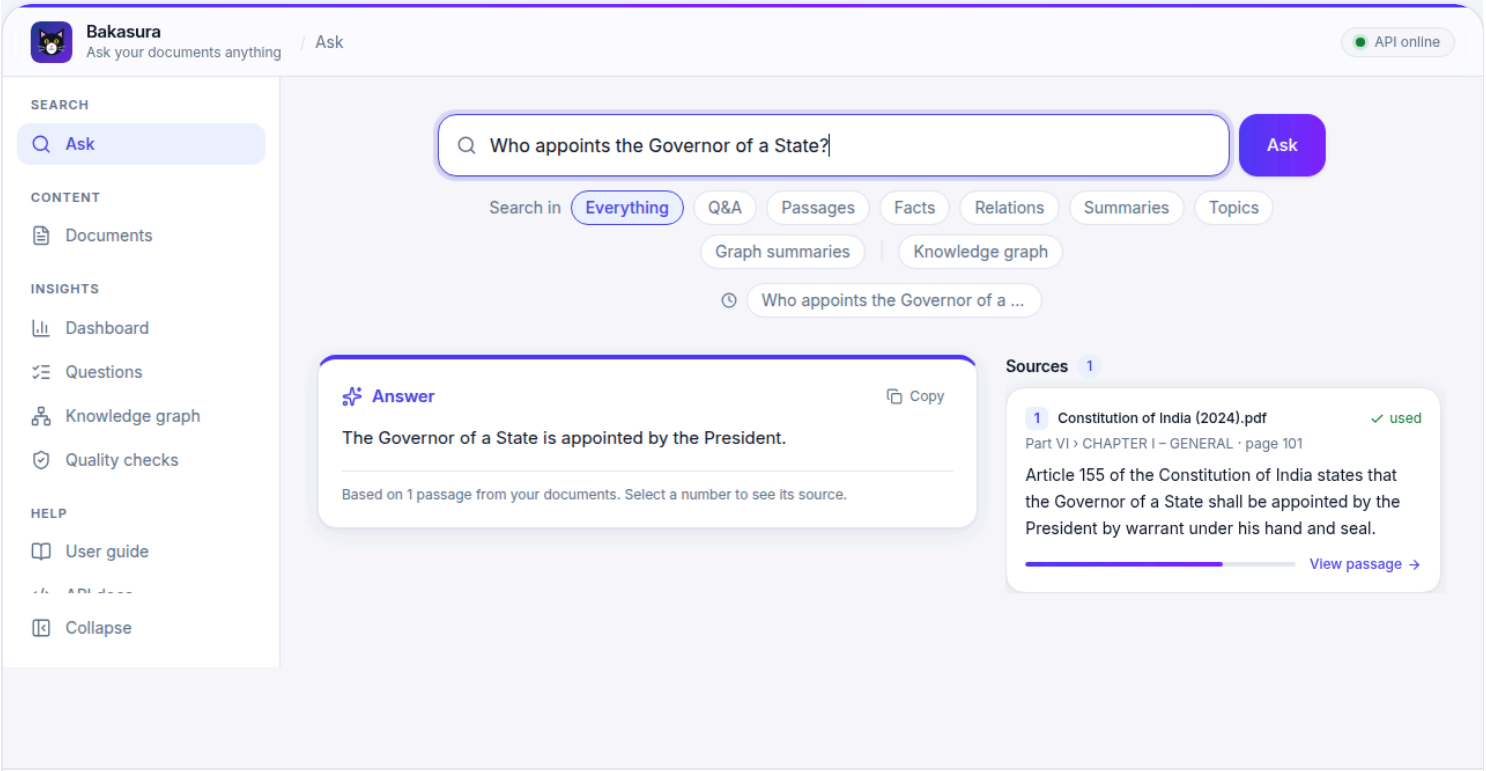
1. **Start the app** with `./start.sh` (Windows: `start.ps1`). It opens the API on port 8000 and this page on port 5173.
2. **Add documents** on the Documents page (PDF, Word or text).
3. **Ask** a question on the Ask page.
4. **Check the answer** by opening the cited passage.
5. **Measure** retrieval on the Dashboard, using the labelled Questions.
6. **Check the LLM’s work** on Quality checks.

The coloured dot at the top right shows whether the API is online. If it says “API offline”, start the app again.

### What happens to a document

- It is cut into **sections** (about half a page, following the document’s headings) and each section into small **chunks** of about 65 words.
- An LLM rewrites vague words in each chunk (“It” becomes “The President”) and writes **Q&A pairs**, **factoids** and **relations** (who does what to whom).
- Everything is turned into vectors and stored. A search matches a small chunk, then the app reads its whole section, so answers have context.

# Ask




The Ask page with a question, a written answer with citations, and source cards

1. Type a question in everyday words, or click one of the suggested questions. Press Enter.
2. The **answer** appears on the left. Numbered chips like **1** are citations: select one to highlight its source card.
3. The **Sources** panel on the right has a card for each passage that helped, rewritten as a short readable answer, with its document, section and page. A bar shows how close the passage is to your question. Cards marked **used** were cited in the answer; other helpful passages can appear without the mark.
4. **View passage** opens the full original text of the section, with buttons to read the previous and next one.

### Filters

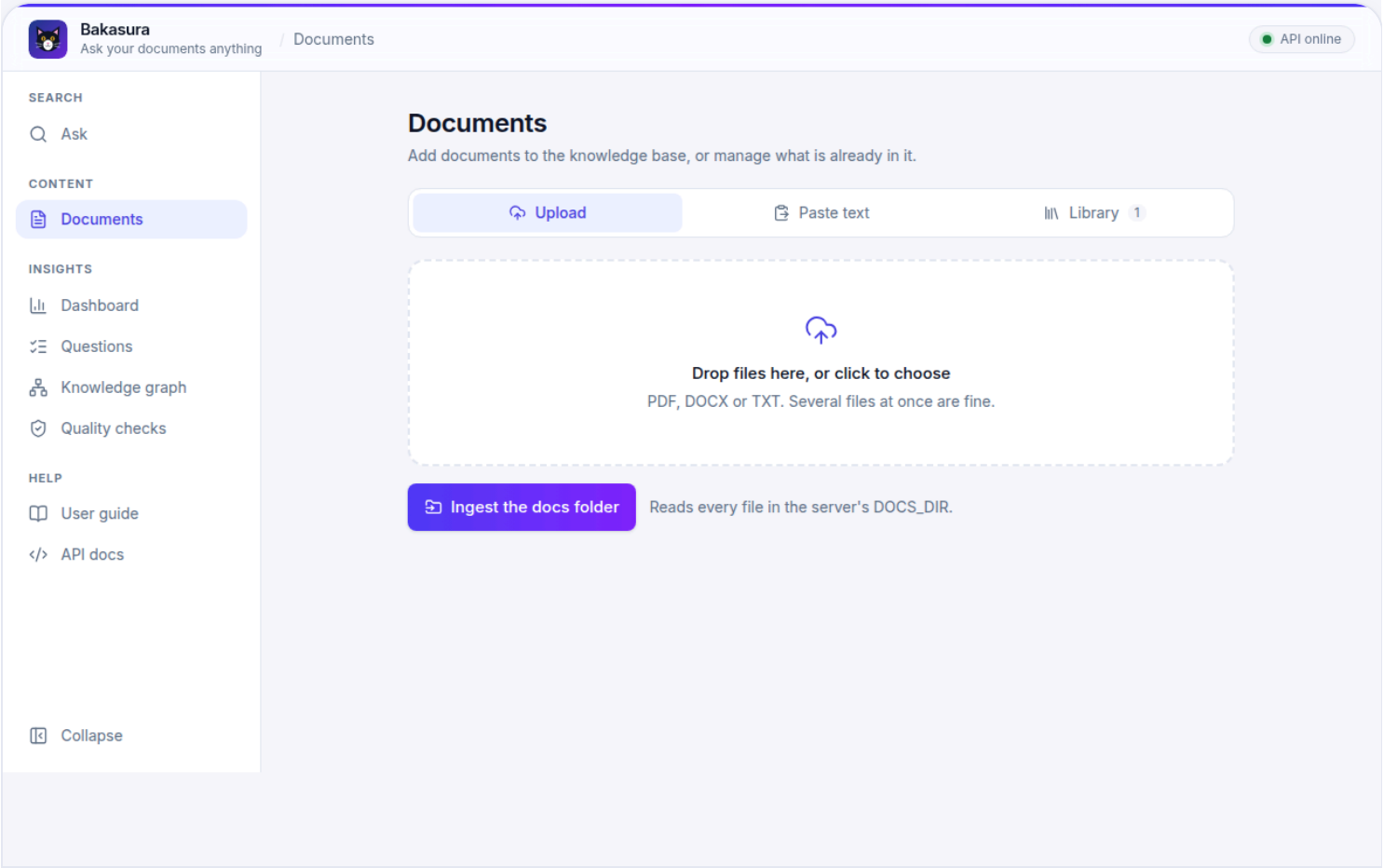
- **Search in** chips limit what is searched: Q&A, Passages, Facts, Relations, Summaries, Topics, Graph summaries. “Everything” (the default) searches the usual types (relations and graph summaries only when you pick them).
- **Knowledge graph** adds passages tied to the entities named in your question. On our test set it neither helped nor hurt, so it starts off. Passages it contributed are tagged “via graph”.

If the answer reads “I couldn’t find this in the documentation”, nothing relevant was retrieved. That is deliberate: the system abstains rather than invent an answer. Try different words, or add the missing

document.

The question is kept in the page address ( `/?q=…` ), so you can bookmark or share an answer.

## Documents




The Documents page with the upload box and the three tabs

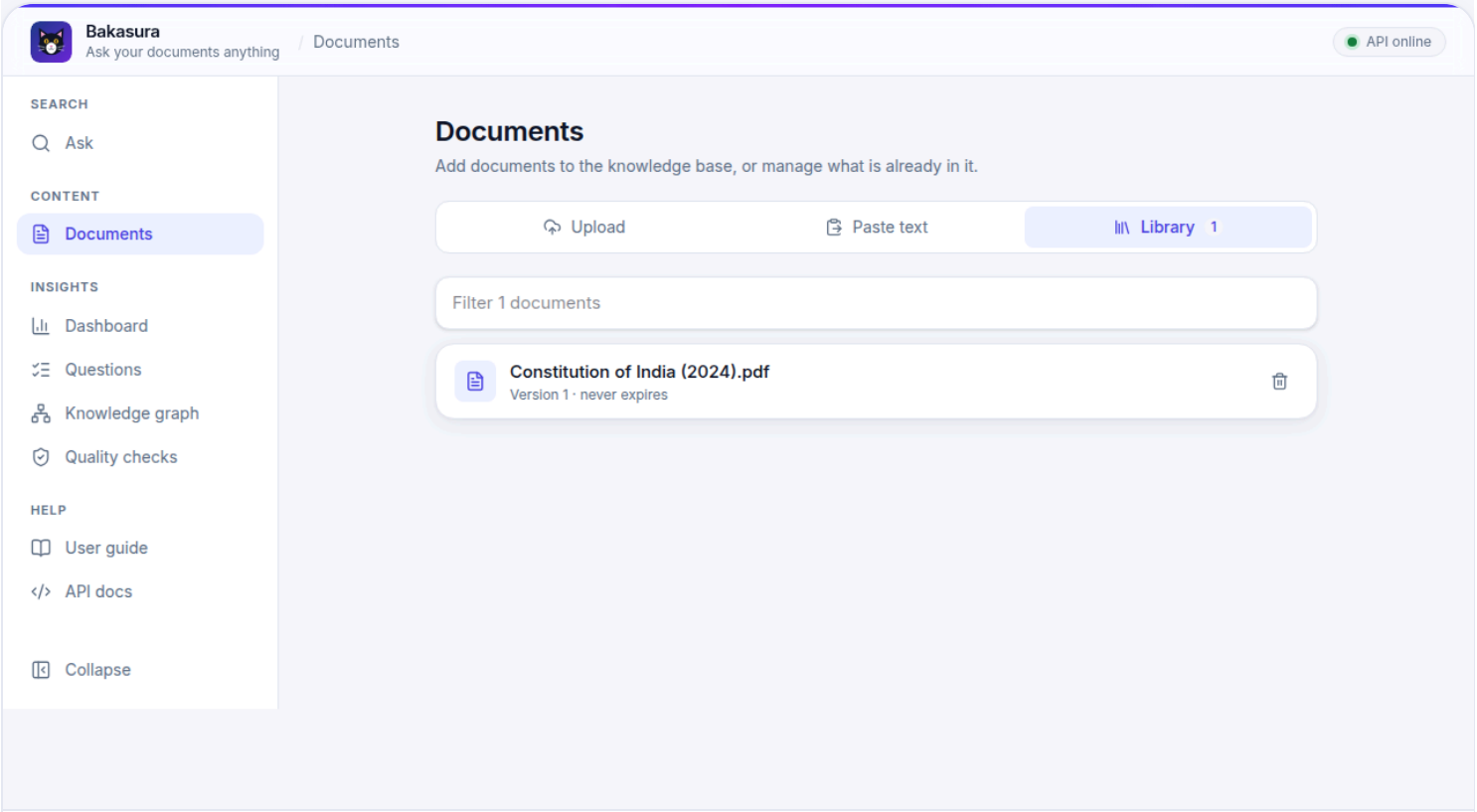
### Add documents

- **Upload**: drag files onto the box or click it. PDF, DOCX and TXT work. You can choose many files at once.
- **Ingest the docs folder** reads every file in the server's `DOCS_DIR` (`data/docs` by default).
- **Paste text**: give a title and paste the text. The title becomes the citation shown with answers.

Ingestion runs in the background and shows progress per file: chunks, Q&A pairs, topics. A large PDF can take tens of minutes because of the LLM provider's rate limit; finished work is cached, so a repeated run is fast.

The same text is never stored twice (it is recognised by a hash and shown as "skipped").

## Library, versions and expiry




The Library tab listing documents with version and expiry

- The **Library** tab lists every document with its **version**, when it was stored and when it expires.
- **Updating**: add a file with the same name or title but changed text. The new version is built first, then the old one is deleted, so a failed update loses nothing. The version number goes up by one.
- **Expiry**: set `DOCUMENT_TTL_DAYS` and documents stored from then on expire after that many days. Expired documents are flushed every `FLUSH_INTERVAL_HOURS` (24 by default) or on demand with `POST /api/documents/flush`. With the default 0, nothing expires.
- **Delete** (the bin icon) removes a document and everything made from it. This cannot be undone.

# Dashboard

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# Dashboard

How well retrieval finds the right passage, and what is in the index.

## Retrieval evaluation

Each question is answered by the section that holds its evidence text. Recall@k: share of right sections in the top k. MRR: 1 / rank of the first right section. nDCG@k: rewards ranking right sections higher.

Retrievers

- [x] BM25
- [x] Base MiniLM (un-tuned)
- [x] Fine-tuned MiniLM
- [x] Hybrid (BM25 + base, RRF)
- [x] Hybrid + cross-encoder rerank
- [x] Base + cross-encoder rerank
- [x] Base + knowledge graph
- [x] Base + knowledge graph + cross-encoder rerank

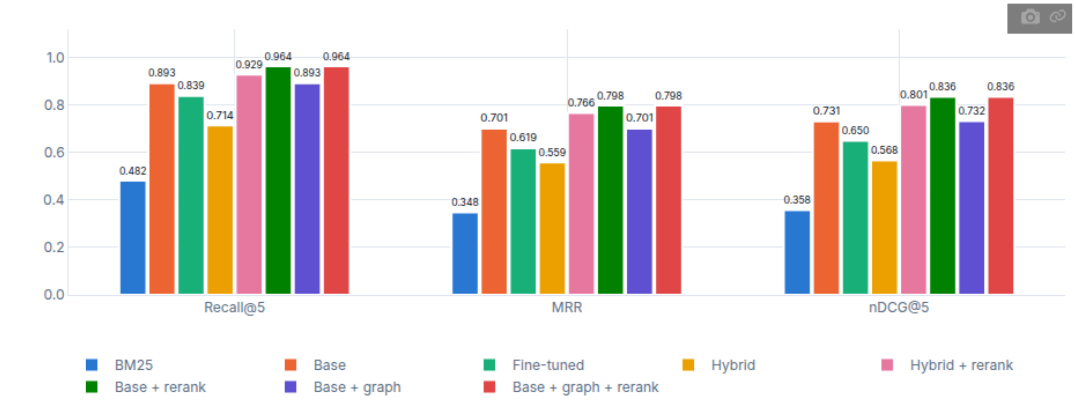
k: 5

Questions: Test split (held-out Parts)

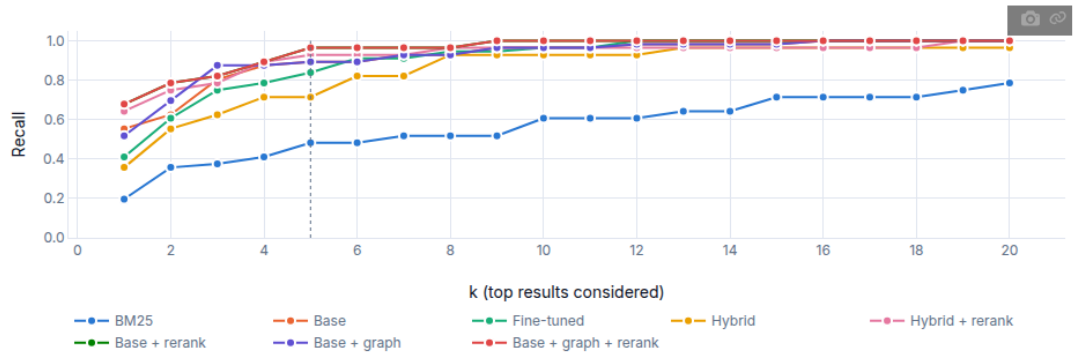
Run evaluation

| Questions scored | Best Recall@5 | Best MRR | Best nDCG@5 |
|---|---|---|---|
| 28 | 0.964 | 0.798 | 0.836 |
| 524 sections, 2358 chunks | Base + rerank | Base + rerank | Base + rerank |

### Retrieval quality by retriever



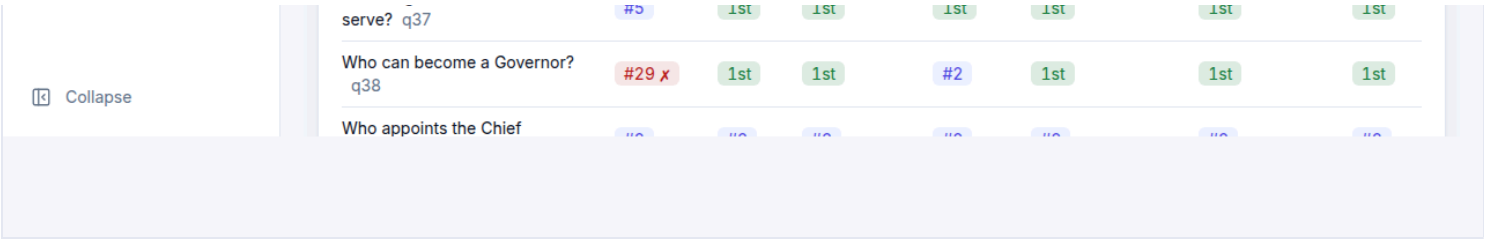

### Recall as k grows




### Per question: where each retriever ranks the right section

Only misses | CSV

| Question | BM25 | Base | Fine-tuned | Hybrid | Hybrid + rerank | Base + rerank | Base + |
|---|---|---|---|---|---|---|---|
| Who is the head of a state government? q35 | #339 ✗ | #3 | #12 ✗ | #25 ✗ | #19 ✗ | #5 | #3 |
| Who appoints the Governor of a state? q36 | #10 ✗ | 1st | #2 | #2 | 1st | 1st | 1st |
| How long does a Governor | | | | | | | |




The Dashboard comparing retrievers by Recall, MRR and nDCG

The Dashboard answers “which way of searching finds the right passage most often?” on the labelled questions.

### Run an evaluation

1. Tick the retrievers, choose **k**, and choose the questions: the **test split** (held-out Parts, the honest number), train, or all.
2. Click **Run evaluation**. The first run embeds the whole index and takes a few minutes; later runs are quick. Progress is shown.
3. Read the comparison table and charts. Download the per-question table as CSV.

### The retrievers

- **BM25**: keyword matching. The baseline.
- **Base MiniLM**: the stock embedding model. **Fine-tuned MiniLM**: the same model trained on this corpus (appears when configured).
- **Hybrid**: BM25 and the base model combined by reciprocal rank fusion.
- **+ cross-encoder rerank**: a second model re-orders the top 30 results. Ask uses base + rerank.
- **+ knowledge graph**: adds graph candidates before ranking.

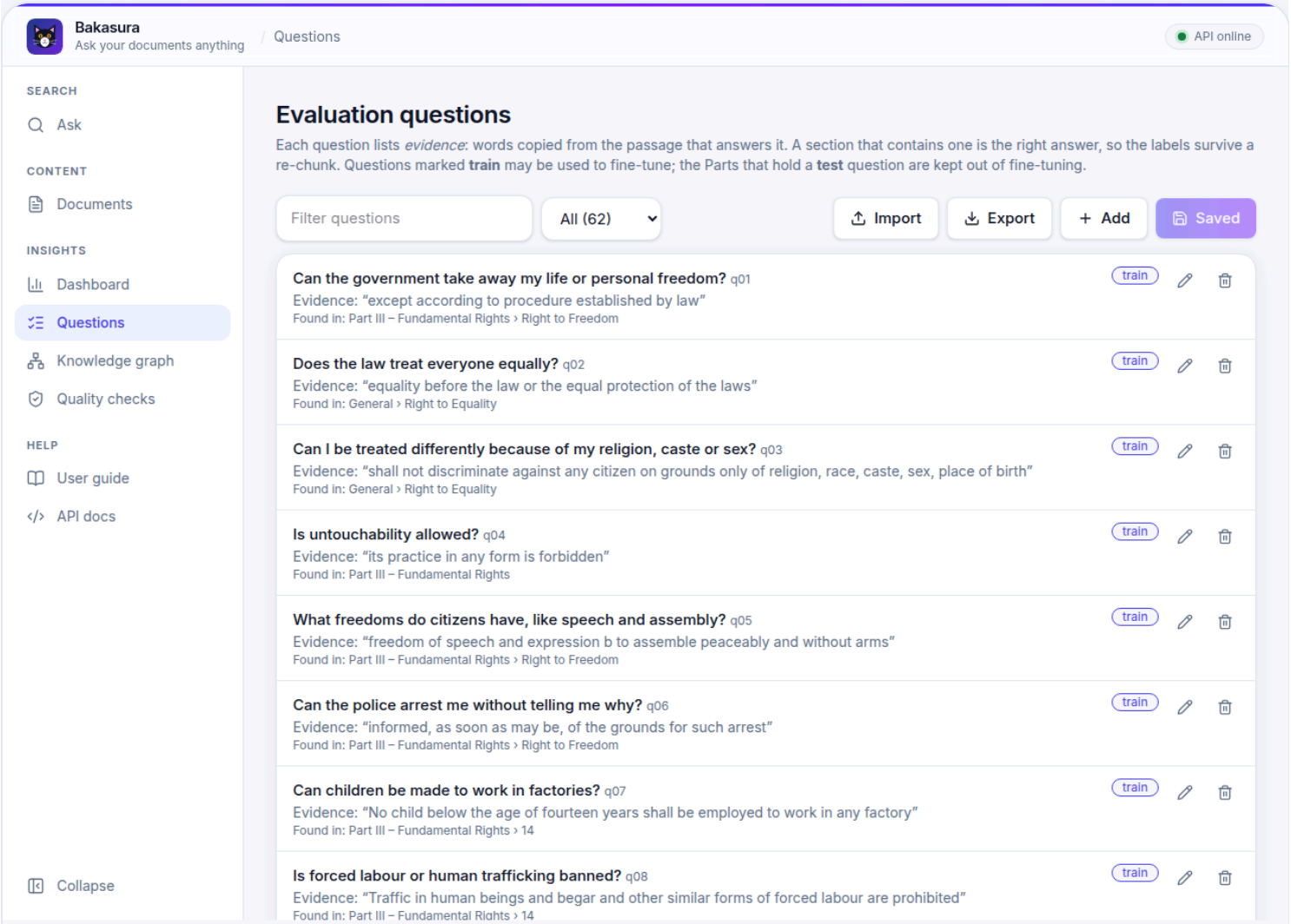
### The metrics

- **Recall@k**: share of questions where the right section is in the top k.
- **MRR**: 1 divided by the rank of the first right section, averaged. Higher means the right answer sits nearer the top.
- **nDCG@k**: like MRR but rewards every right section and discounts lower ranks.

The per-question table shows the rank each retriever gave the right section; misses are highlighted. The index statistics show how many points of each type exist and how big the chunks are.

With few test questions, a difference of one or two questions is noise. Treat small gaps as ties.

# Questions




The Questions page listing labelled evaluation questions

This is the answer key for the Dashboard. Each question lists **evidence**: a phrase copied from the passage that answers it. Any section containing the phrase counts as correct, so the labels survive re-chunking.

1. Click **Add**. Give an id, the question in everyday words, and one evidence phrase per line.
2. Choose **test** or **train**. Train questions may be used to fine-tune the model; the Parts of the document holding a test question are kept out of training so the result is not inflated.
3. Under each question the page shows which sections contain its evidence. Not found in any section means a typo or a changed document: an evaluation would leave it out.
4. Click **Save changes**. **Import** and **Export** read and write the JSON file.

The most honest test questions are written by someone who has not read the documents.

## Knowledge graph

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# Knowledge graph

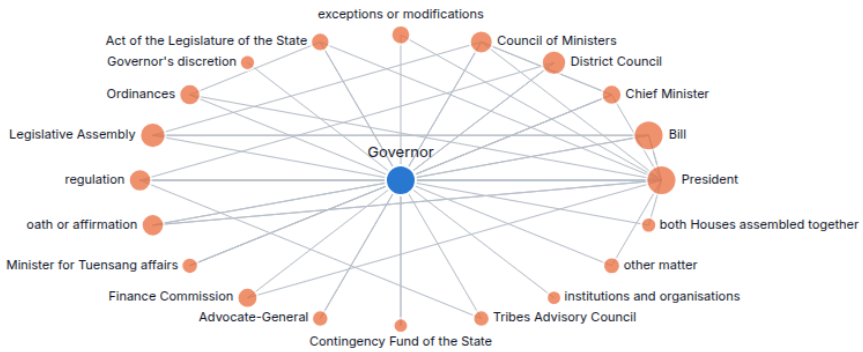
Entities and relations the LLM found in each passage. Spellings are merged ("the Governors" and "Governor" are one entity). Pick an entity to see how it connects; every relation links back to the passage it came from.

| Entities | Relations | Documents | Rebuild the graph |
|---|---|---|---|
| 5,212 | 5,416 | 1 | Runs automatically after each ingest. |

Find an entity, e.g. Governor, High Court, Lok Sabha

## Governor

20 of 157 neighbours shown, the most connected first. Select an entity to move to it.




### Relations of Governor (40+)

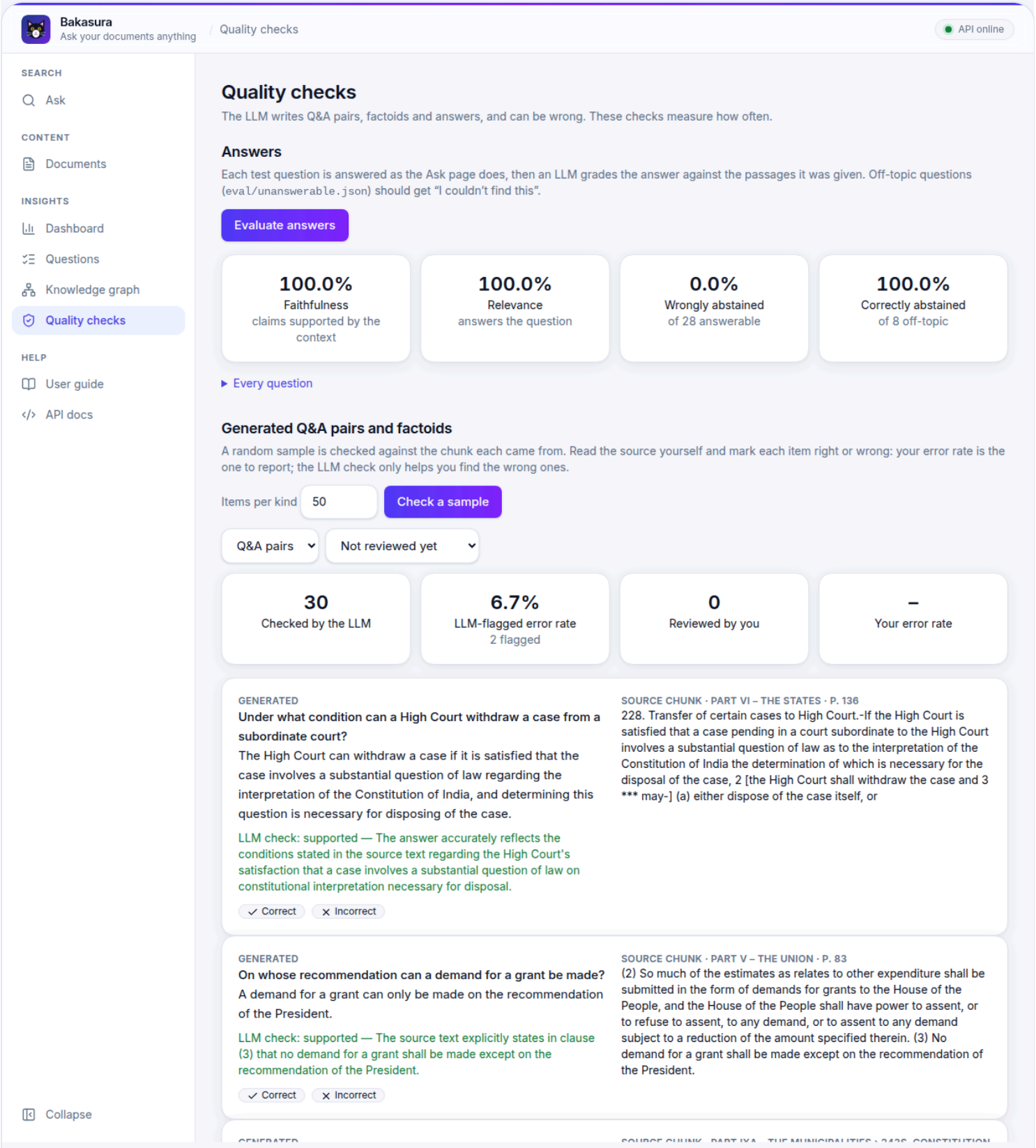
| Relation | Where it was stated | |
|---|---|---|
| Governor makes rules for transaction of the business of the Government of the State | Part VI › Conduct of Government Business · page 105<br>(3) The Governor shall make rules for the more convenient transaction of the business of the Government of the State, and for the allocation among... | Passage → |
| Governor administers autonomous district | Sixth Schedule › 1 * * * * · page 334<br>1 19. Transitional provisions.-(1) As soon as possible after the commencement of this Constitution the Governor shall take steps for the... | Passage → |
| Governor takes steps for constitution of District Council | Sixth Schedule › 1 * * * * · page 334<br>1 19. Transitional provisions.-(1) As soon as possible after the commencement of this Constitution the Governor shall take steps for the... | Passage → |
| Governor may entrust functions to District Council | Sixth Schedule › 1 * * * * · page 322<br>(2) The Governor may, with the consent of any District Council, entrust either conditionally or unconditionally to the District Council or to the District... | Passage → |
| Governor requires consent of District Council | Sixth Schedule › 1 * * * * · page 322<br>(2) The Governor may, with the consent of any District Council, entrust either conditionally or unconditionally to the District Council or to the District... | Passage → |
| Governor makes a report to President | Part XXI › SPECIAL PROVISIONS] · page 268<br>(2) The Governor shall annually, or whenever so required by the President, make a report to the President regarding the administration of the Hill Area... | Passage → |
| Governor requires instructions from President | Procedure in Financial Matters · page 128<br>Provided that the Governor shall not, without instructions from the President, promulgate any such Ordinance if- (a) a Bill containing the same provisions... | Passage → |
| Governor reserves for consideration of President | Part VI – The States · page 122<br>200. Assent to Bills.-When a Bill has been passed by the Legislative Assembly of a State or, in the case of a State having a Legislative Council,... | Passage → |
| Governor reserves for consideration of President | Sixth Schedule › 1 * * * * · page 317<br>(2) All laws made under paragraph 3 or under this paragraph shall in so far as they relate to matters specified in List III of the Seventh Schedule, be... | Passage → |
| Governor shall make a report to President | Part C › FIFTH SCHEDULE · page 307<br>Report by the Governor 5 *** to the President regarding the administration of | Passage → |

The Knowledge graph page showing an entity, its neighbours and its relations

The graph is built from the relations the LLM found in each chunk (“Governor — appoints — Chief Minister”). Different spellings of one thing merge into one entity, and every relation remembers the chunk that stated it.

- A graph with thousands of entities is unreadable as one picture, so this page shows **one entity at a time**: search for it, or click a bar in the most-connected chart.
- The network shows the entity in the centre and its closest neighbours around it. **Click a neighbour** to move there.
- The table lists the relations with the section, page and a link to the source passage. Always check the source: the relations are only as good as the LLM’s reading.
- **Rebuild** recreates the graph from the stored relations. It also happens automatically after every upload or deletion.

# Quality checks




The Quality checks page with answer scores and a generated item next to its source

An LLM wrote the Q&A pairs, factoids and answers, so some are wrong. This page measures how many.

#### Answers

Click **Evaluate answers**. Every test question is answered as Ask would, and a second LLM call grades each answer against the passages it was given:

- **Faithfulness**: every claim is supported by the passages.
- **Relevance**: the answer addresses the question.
- **Wrongly abstained**: questions that have an answer but the system said "I couldn't find this".
- **Correctly abstained**: off-topic questions (listed in `eval/unanswerable.json`) that were refused, as they should be.

#### Generated Q&A pairs and factoids

1. Choose how many random items to check and click **Check a sample**. Each costs one LLM call.
2. For each item the page shows the generated text beside the chunk it came from, and the LLM's verdict.
3. Read the source and click **Correct** or **Incorrect** (click again to clear). Use "Flagged by the LLM" to look at the likely errors first.
4. The tiles show the LLM-flagged error rate, your own error rate and how often you agree. **Report your own rate**: it is the trustworthy one, and a sample of 50 to 100 is enough to be useful.

## Running and configuring

#### Start and stop

- `./start.sh` starts Qdrant (Docker), the API and this page. `./stop.sh` stops the API and page; `./stop.sh --all` also stops Qdrant. Your data is kept.
- **Without Docker** (for example on Windows): set `QDRANT_PATH=data/qdrant` in `.env`. The index is then a folder inside the project. To copy an existing index into it, run `uv run python -m capstone.export_index data/qdrant` once. The folder can be zipped and moved to another computer.
- On Windows use `start.ps1` and `stop.ps1`.

#### Settings that matter

Everything is in `.env` (see `.env.example`). The ones you are most likely to change:

- `GROQ_API_KEY`: your LLM key. Required.

- `COLLECTION`: the name of the index. **Use a new name** whenever you change the embedding model, then ingest again.
- `DOCUMENT_TTL_DAYS`, `FLUSH_INTERVAL_HOURS`: expiry of new documents.
- `GRAPH_SEARCH`: use the knowledge graph in every search (off by default).
- `ANSWER_MIN_SCORE`: how similar a passage must be to count. Raise it to abstain more often.

## Troubleshooting

- **"API offline"**: the API is not running or still loading its models (the first start downloads them). Wait a minute, then run `./start.sh` again.
- **"GROQ_API_KEY" missing**: copy `.env.example` to `.env` and put your key in it.
- **Ingestion is slow, or some files fail**: the LLM provider limits how much text it will write per minute. Run it again; cached work is not repeated.
- **The Dashboard has no graph retrievers**: the graph is built after the first ingest. Use Rebuild on the Knowledge graph page.
- **Answers say "I couldn't find this" for something that is in the documents**: use words closer to the document, or lower `ANSWER_MIN_SCORE` a little.
- **A question is "Not found in any section"** on the Questions page: its evidence phrase is not in the current text. Fix the phrase or remove the question.
- Technical details for developers are on the API docs page.